

not something you do because you have to. Tinkering is work you do for fun.

Hackers embrace the playfulness of tinkering, but here's the mischief in their creed: just because the work is fun does not mean it is unimportant. 'Playing', in the hacker sense of the word, is not just a way to stay entertained. It is an attitude toward innovation that champions gamesmanship and admires intellect applied with competitive vigour and flair. In chess, the grandmaster and the goat each play with the same sixteen pieces. But in the hands of the former, the game becomes an object of beauty and raw intellectual force. In the same way, the gifted tinkerer can rearrange the already existing engine parts or snippets of computer code in a way that creates something utterly new and potentially transformative.

For hackers, the logical frame of mind required for programming spills over into more commonplace activities. You could ask hackers a question and sense their mental accumulators processing bits until they came up with a precise answer to the question you asked. Marge Saunders would drive to the Safeway supermarket every Saturday morning in her Volkswagen and on her return would ask her husband, "Would you like to help me bring in the groceries?" Bob Saunders would reply, "No." Stunned, Marge would drag in the groceries herself. After this occurred a few times, she exploded, hurling curses at him and demanded an explanation on why he didn't help her.

"That's a stupid question to ask," he said. "Of course I won't like to help you bring in the groceries. If you ask me if I will help you bring them in, that's another matter." It was as if Marge had submitted a program into the TX-0, and the program, as programs do when the syntax is improper, had crashed. It was not until she debugged her question that Bob Saunders would allow it to run successfully on his own mental computer.

Hacker ethic

Wikipedia accurately explains the 'hacker ethic' as a generic phrase that describes the moral values and philosophy that are standard in the hacker community. The early hacker culture and resulting philosophy originated at the Massachusetts Institute of Technology (MIT) in the 1950s and 1960s. The term 'hacker ethic' is attributed to journalist Steven Levy, as described in his book titled 'Hackers: Heroes of the Computer Revolution', written in 1984. The guidelines of the hacker ethic make it easy to see how computers have evolved into the personal devices we know and rely upon today.

The hacker ethic was a "...new way of life, with a philosophy, an ethic and a dream." However, the elements of the hacker ethic were not openly debated and discussed; rather they were accepted and silently agreed upon.

Free and Open Source Software (FOSS) has evolved

from the hacker ethics that Levy described. The hackers who stay true to the hacker ethics—especially the Hands-On Imperative—are usually supporters of the free and open source software movement.

The general tenets of the hacker ethic are:

Access to computers—and anything that might teach you something about the way the world works—should be unlimited and total. Always yield to the Hands-On Imperative! Hackers believe that essential lessons can be learned about the systems—about the world—from taking things apart, seeing how they work, and using this knowledge to create new and even more interesting things. They mostly resent any person, physical barrier, or law that tries to keep them from doing this.

All information should be free. If you don't have access to the information you need to improve things, how can you fix them? A free exchange of information, particularly when the information is in the form of a computer program, allows for greater overall creativity.

Mistrust authority—promote decentralisation. The best way to promote this free exchange of information is to have an open system, with no boundaries between hackers and a piece of information, or an item of equipment that they need in their quest for knowledge, and their time on-line. The last thing they need is a bureaucracy, whether in the corporate world, in government, or at university. Bureaucracies are flawed systems, dangerous in that they cannot accommodate the exploratory impulses of true hackers. Bureaucrats hide behind arbitrary rules (as opposed to the logical algorithms by which machines and computer programs operate): they invoke those rules to consolidate their own power, and perceive the constructive impulse of hackers as a threat.

Hackers wish to be judged by their hacking, not bogus criteria such as degrees, age, race, or position. Hacker cultures are meritocracies, where positions are based on demonstrated knowledge and achievements. Hackers care less about people's superficial characteristics than they do about their potential to advance the general state of hacking, to create new programs to admire, to talk about new features in the system, etc.

You can create art and beauty on a computer. Hackers deeply appreciate innovative techniques that allow programs to perform complicated tasks with a few instructions. A program's code is considered 'beautiful' in its own right, having been carefully composed and artfully arranged. Learning to create programs that used the least amount of space almost became a game between early hackers.

Computers can change your life for the better. This belief is subtly manifest. Rarely will hackers try to impose their view of the myriad advantages of the computer way of knowledge to an outsider. Yet, this